EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences

LABORATORY REPORT

CHE109 - ENGINEERING CHEMISTRY

Experiment No: 02

EXPERIMENT TITLE

Determination of the Molar Mass of an Unknown Diprotic Acid using NaOH

SUBMITTED TO

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DATE OF EXPERIMENT	**30 September 2026**
DATE OF SUBMISSION	**07 October 2026**

Experiment 02: Determination of the Molar Mass of an Unknown Diprotic Acid using NaOH

Objective

The aim of this experiment was to determine the molar mass of an unknown diprotic acid (H_2A) by titrating a 10.0 mL aliquot with standardized 0.100 M sodium hydroxide (NaOH), using phenolphthalein as the indicator.

Introduction and Principle

Diprotic acid can donate two protons per molecule. In this experiment, the unknown acid H_2A reacts with sodium hydroxide in a 1:2 mole ratio. The average volume of standardized NaOH required to reach the phenolphthalein endpoint is used to calculate the molarity of H_2A. The acid concentration and the recorded mass of 6.3 g are then used to determine the experimental molar mass.

Reaction

$2NaOH(aq) + H_2A(aq) \rightarrow Na_2A(aq) + 2H_2O(l)$

Chemicals and Apparatus

Chemicals: Diprotic acid (H_2A), sodium hydroxide (NaOH), phenolphthalein indicator, distilled water.

Apparatus: Volumetric flask, conical flask, burette, pipette, funnel, wash bottle, burette stand and clamp.

Procedure

a) Preparation of 0.1 M, 250 mL NaOH solution

Weigh 1.000 g of NaOH using the formula below (molar mass of NaOH = 40 g/mol).
$W = SMV/1000 = 0.1 \times 40 \times 250/1000 = 1.000$ g
Transfer the NaOH to a 250 mL volumetric flask. Add a small amount of distilled water and shake until the solute dissolves. Add more distilled water up to the 250 mL mark.

b) Determination of the concentration of H_2A

1. Fill the burette with prepared NaOH and carefully set up the burette on a stand.
2. Record the initial burette reading.
3. Pipette 10 mL of H_2A into a conical flask with the help of pipette.
4. Add two drops of phenolphthalein indicator.
5. Titrate until the last drop of NaOH solution leaves a permanent **pink** colour in the solution and records the final burette reding.
6. Repeat the titration two more times.

Experimental Data

Determination of the concentration of H_2A solution with NaOH solution.

Observation	Initial burette reading (mL)	Final burette reading (mL)	NaOH used (mL)
1	10.5	20.6	10.1
2	31.1	41.4	10.3
3	22.8	33.0	10.2
Average			10.2

Calculations

According to the reaction, $2 \times M(acid) \times V(acid) = M(base) \times V(base)$.

$M(H_2A) = [M(NaOH) \times V(NaOH)] / [2 \times V(H_2A)] = [0.100 \times 10.2] / [2 \times 10.0] = 0.051$ mol/L

Molar Mass and Percentage Error

For 1000 mL of H_2A solution, n = 0.051 mol. Molar mass = weight taken / amount (mol). Percentage error = |theoretical − experimental| / theoretical × 100.

Experimental molar mass = 6.3 g / 0.051 mol = 123.53 g/mol; percentage error = |126.07 − 123.53| / 126.07 × 100 = 2.02%.

Result

The unknown diprotic acid had an experimental molar mass of 123.53 g/mol. The average NaOH volume was 10.2 mL, giving an H_2A concentration of 0.051 mol/L.

Discussion / Error Analysis

The NaOH volumes were 10.1, 10.3, and 10.2 mL, with an average of 10.2 mL. Their close agreement indicates consistent titration readings. Using the recorded 6.3 g mass, the calculated molar mass was 123.53 g/mol, compared with the theoretical value of 126.07 g/mol. The percentage error was 2.02%. Small differences may result from endpoint judgement, burette parallax, air bubbles, or measurement and transfer errors. Accuracy can be improved by reading the meniscus at eye level, removing air bubbles, rinsing glassware with the solutions used, and adding NaOH dropwise near the endpoint.

Answers to Practice Questions

1. **What are monoprotic, diprotic and triprotic acids?**

A A monoprotic acid donates one proton per molecule, a diprotic acid donates two, and a triprotic acid donates three. Examples include HCl, H_2A, and H_3PO_4, respectively.

2. **Why is phenolphthalein used as a suitable indicator for this titration?**

A Phenolphthalein is colorless in acidic solution and turns pink in slightly basic solution. Its visible colour change provides a clear endpoint as NaOH is added to the weak diprotic acid.

3. **Discuss the precautions of the above experiment.**

A Use clean glassware and rinse the burette with NaOH and the pipette with the acid solution before use. Ensure the burette tip contains no air bubbles and check for leaks. Read the meniscus at eye level, add only a few drops of indicator, swirl the flask continuously, and add NaOH slowly near the endpoint. Record initial and final readings carefully and repeat the titration for consistent results.